

Example

Let $f : A \longrightarrow B$ be defined by $f(x) = 2x + 3$. Evaluate the following

1. $f(0)$
2. $f(1)$
3. $f(-3)$
4. $f(a)$

Example

A function is defined by $f : x \longrightarrow \frac{3x-1}{x-3}$ on $\{-2, -1, 0, 1, 2, \}$.

1. Find the range of f
2. What values of x make the function undefined?



Example

State the largest possible domain of the functions

$$1. f(x) = \frac{1-2x}{\sqrt{2x-1}}$$

$$2. f : x \longrightarrow \frac{x+5}{x-3}$$

$$3. f(x) = 1 - 2x$$

$$4. g : x \longrightarrow \sqrt{2x-1}$$

$$5. g : x \longrightarrow \frac{x}{\sqrt{2x-1}}$$

$$6. g(x) = \sqrt{1-x^2}$$

$$7. f(x) = \sqrt{5x-2}$$

$$8. f(x) = \frac{\sqrt{x-2}}{x^2-4}$$



Examples

$$1. \lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$$

$$2. \lim_{x \rightarrow 4} (2x + 3)$$

$$3. \lim_{x \rightarrow 1} \left\{ \frac{x^2-1}{x+1} \right\}$$

$$4. \lim_{x \rightarrow 2} \left\{ \frac{x^2-4}{x-2} \right\}$$

$$5. \lim_{x \rightarrow 4} \left\{ \frac{x^2-7x+10}{x^2-4} \right\}$$

$$6. \lim_{x \rightarrow 1} \left(\frac{x^3-1}{x-1} \right)$$

$$7. \lim_{x \rightarrow 9} \left\{ \frac{9-x}{3-\sqrt{x}} \right\}$$

$$8. \lim_{x \rightarrow 3} \left\{ \frac{9-x^2}{4-\sqrt{x^2+7}} \right\}$$



Examples

Compute the following limits

$$1. \lim_{x \rightarrow 2} (x + 3)(x^2 - 5)$$

$$2. \lim_{x \rightarrow 2} f(x) \text{ for } f(x) = \begin{cases} 1 - x, & x \leq 2 \\ 4, & x > 2 \end{cases}$$

$$3. \lim_{x \rightarrow 0} \frac{|x|}{x}$$

$$4. \lim_{x \rightarrow 1} f(x) \text{ for } f(x) = \begin{cases} x + 1, & x \neq 1 \\ \pi, & x = 1 \end{cases}$$



5. Determine the existence or otherwise of $\lim_{x \rightarrow 4} f(x)$ where

$$f(x) = \begin{cases} \sqrt{x-4}, & \text{if } x > 4 \\ 8-2x, & \text{if } x < 4 \end{cases}$$

6. Evaluate $\lim_{x \rightarrow 1} f(x)$ where $f(x) = \begin{cases} \frac{x-1}{x^2-1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$

* Show that $\lim_{x \rightarrow 0} |x| = 0$



Example

Evaluate the following limits

$$1. \lim_{x \rightarrow \infty} \left(\frac{5x^2 - 1}{2x^2 + 1} \right)$$

$$2. \lim_{x \rightarrow \infty} \left(\frac{7x}{x - 4} \right)$$

$$3. \lim_{x \rightarrow \infty} \left(\frac{2x^3 - 7x + 3}{x^3 + 2} \right)$$

$$4. \lim_{x \rightarrow \infty} \left(\frac{3x}{x + 2} \right)$$



Examples

1. Show that $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$
2. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$
3. Compute $\lim_{x \rightarrow 0} \sec x$

Example

Evaluate the following limits

1. $\lim_{x \rightarrow 0} (\cos x + \sin x)$
2. $\lim_{x \rightarrow 0} \left(\frac{2 \tan x - x}{\sin 2x} \right)$
3. $\lim_{x \rightarrow 0} \left(\frac{\sqrt{2} - \sin x}{\cos x} \right)$
4. $\lim_{x \rightarrow 0} \left(\frac{\sin 5x}{x} \right)$
5. $\lim_{x \rightarrow 0} \left(\frac{\cot x}{\cot 2x} \right)$



Definition

A function $f(x)$ is continuous on an interval (a, b) if it is continuous for all x in the interval.

Examples

1. Show that $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$
2. Determine the continuity of $f(x) = \frac{x^2 - 4}{x - 2}$ at $x = 1$ and at $x = 2$



Remark

1. Every polynomial function is continuous for all real x .
2. A rational function is continuous for all real x except those values of x for which the denominator equals zero.
3. Every exponential function is continuous for all real x .
4. The natural logarithmic function, $\ln |x|$ is continuous for all real x in its domain.
5. A radical function, or rational exponent function is continuous for all real x in its domain.



Example

Determine the continuity of the given function at the point indicated

1. $f(x) = 3x - 2$; $x = -2$

2. $f(x) = 2x + 6$; $x = -3$

3. $h(x) = x^2 - x - 6$; $x = 3$

4. $h(x) = 2x^3x^2 - 5$; $x = 2$

5. $g(x) = \frac{x+1}{x-3}$; $x = 3$

6. $g(x) = \frac{x-5}{x+2}$; $x = -2$

7. $f(x) = \frac{x^2-25}{x-5}$; $x = 5$

8. $f(x) = \frac{x^2-25}{x-5}$; $x = 0$



$$9. f(x) = \begin{cases} x + 2, & x \leq 1 \\ x^2 + 3, & x > 1 \end{cases} \quad x = 1$$

$$10. f(x) = \begin{cases} x - 3, & x \leq 0 \\ x^2 + x - 3, & x > 0 \end{cases} \quad x = 0$$



Example

Determine the intervals where the following functions are continuous;

$$1. f(x) = \frac{2x+1}{(x-1)(2x+1)}$$

$$2. f(x) = (1.12)^x$$

$$3. f(x) = \sqrt{x-7}$$

$$4. f(x) = 7x^2 - 3.2x + 0.5$$

$$5. f(x) = \frac{x+1}{(x+1)(x-3)}$$

$$6. f(x) = 4x^3 - 2x^2 + 1.3x - 5$$

$$7. f(x) = \sqrt{2x+3}$$

$$8. f(x) = e^{-0.25x}$$

$$9. f(x) = -2.1 + \ln x + 3$$

$$10. f(x) = \frac{3x-7}{2x+1}$$

$$11. f(x) = \frac{2x-3}{(2x-3)(x+4)}$$



Example

Find from first principles the derivative of the following;

1. $f(x) = x^2$

2. $f(x) = x^2 + x$

3. $f(x) = 5x$

4. $f(x) = 8$

5. $f(x) = \frac{1}{x^2}$

Example

Differentiate the following functions from first principles

1. $f(x) = 2x^3 + 3$

2. $f(x) = 2x^2 + 4x - 3$

3. $f(x) = 3x^2 + 1$

4. $f(x) = \sqrt{x}$

5. $f(x) = (2x + 3)^2$



Product Rule of Differentiating

If $h(x) = f(x)g(x)$, where $f(x)$ and $g(x)$ are differentiable functions, then $h'(x) = f'(x)g(x) + f(x)g'(x)$.

Examples

1. Differentiate $h(x) = 3x^3(x^4 + 2)$
2. Find the derivative of the following functions
 - a. $f(x) = (2x^2 + 4x + 5)(5x - 4)$
 - b. $y = \sqrt{x}(3x^3 - 4x^2 + 8)$
 - c. $y = (6x^{4/3} + 2x)(3x^{5/3} + 4x - 1)$



Quotient Rule of Differentiating

If $h(x) = \frac{f(x)}{g(x)}$ where $g(x) \neq 0$, and f and g are differentiable functions the,

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}; \quad g(x) \neq 0$$

Example

Find the derivative of the following functions;

1. $f(x) = \frac{x+3}{x-2}$

2. $y = \frac{x^4-3x}{x^2+1}$

3. $y = \frac{(2x+1)(3x-2)}{x+1}$

4. $y = \frac{1+x}{1-x^2}$

5. $y = \frac{x^2-1}{2x^2+1}$

6. $y = \frac{(x-3)^2}{x^2+1}$

7. $y = \frac{3x-1}{\sqrt{x^2+1}}$

8. $y = \frac{x}{1+x^2}$

9. $y = \frac{x^2-1}{x^3+1}$



The Chain Rule

This rule is used to differentiate composite functions that have the form $h(x) = f(g(x))$. If $y = f(u)$ and $u = g(x)$ then $y = f(g(x))$. Therefore $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ provided that $\frac{dy}{du}$ and $\frac{du}{dx}$ exist.

Example

Find the derivatives of the following functions using the chain rule.

1. $f(x) = (5x^3 + 3x)^4$

2. $f(x) = (x^2 + 1)^{15}$

3. $f(x) = \sqrt[3]{2x - 4}$

4. $f(x) = \frac{5}{(2x-3)^2}$

5. $g(x) = (x + 3)^5$

6. $y = (2x^2 - 5x)^4$



The Generalised Power Rule

If u is a differentiable function of x and n is any real number with $f(x) = \{u(x)\}^n$, then $f'(x) = n\{u(x)\}^{n-1} \cdot u'(x)$. In simple terms, if $y = (ax + b)^n$, then

$$\frac{dy}{dx} = n(ax + b)^{n-1} \cdot \frac{d}{dx}(ax + b)$$

Example

Find the derivatives of the following functions

1. $y = (x^2 + 2)^5$
2. $y = (3x^4 - 5)^7$
3. $y = (2x - 5)^{-4}$
4. $y = (3x + 4)^5$



Example

Compute the derivatives of the following functions

1. $f(x) = 2 \ln x$

2. $f(x) = \ln x^3$

3. $f(x) = 7 - 4 \ln x$

4. $f(x) = \ln(x^4 - 2x)$

5. $y = \ln \sqrt{6x - 1}$

6. $y = (\ln x)^3$

Example

Determine the derivative of the following functions

1. $y = 4x^3(\ln x)$

2. $y = (\ln x)^2$

3. $y = \frac{3x^5}{\ln x}$

4. $y = 12x^3(\ln x^2)$



Derivative of the General Logarithmic Function

The general logarithmic function is the function $f(x) = \log_b x$, where $b > 0$, $b \neq 1$. The function $f(x) = \log_b x$ can be written as $f(x) = \frac{\ln x}{\ln b}$ via the change of base formula. Hence $f'(x) = \frac{1}{\ln b} \left(\frac{1}{x} \right)$

In general, if $f(x) = \log_b \{g(x)\}$ then $f'(x) = \frac{1}{\ln b} \cdot \frac{g'(x)}{g(x)}$



Derivative of the Exponential Function

The function $h(x) = e^{f(x)}$, where $e = 2.718$ is a constant is called the exponential function. If f is a differentiable function of x , then $h'(x) = e^{f(x)} \cdot f'(x)$.

Examples

Differentiate the following functions with respect to x

1. $y = x^2 e^x$

2. $y = \frac{x^2}{e^x}$

3. $f(x) = \log_e e^x$

4. $h(x) = e^{3x-3}$

5. $g(x) = e^{6x - \frac{1}{2}x^6}$



Examples

Determine the derivative of the following functions

1. $y = 10^x$

2. $y = \frac{3^x}{4^x}$

3. $y = 10^{x+3}$

4. $y = 5^{x^2}$



Example

Find the derivative of these functions

1. $y = 5^{9x-5}$

2. $y = 3^{\ln x + 5}$

3. $y = 2^{9-x}$

4. $y = 4^{\sqrt{x}}$

5. $y = 4^{x-3}$

6. $y = 5^{x^2}$



Example

Differentiate the following with respect to x .

1. $y = \sin x^2$

2. $y = \sin (x^2 - 3x + 1)$

3. $y = \cos (2x^3 + x^2 - 1)$

4. $y = \cos (4x^3)$

5. $y = \sin (3x^2 + 1)$

6. $y = \tan 6x^2$



Find the derivative of the following functions with respect to x

1. $y = \tan^2 x$

2. $y = 2 \cos^2 (x^2 + 1)$

3. $y = \sin^3 x$

4. $y = 3 \tan^4 x$

5. $y = x \sin 3x$

6. $y = x^3 \cos 2x$

7. $y = \cos x \sin^3 x$

8. $y = 2x^3 \cos 3x$

9. $y = x^4 \cos x$

10. $y = \cos (x^2 - x)$



Example

Use implicit differentiation to determine $\frac{dy}{dx}$ for the following functions

1. $y^3 - 4x^2 - 7 = 0$

2. $x^{1/3} + y^{1/3} - 10 = 0$

3. $4e^y = x^2$

4. $2x + 1 = \sqrt{2 - y^2}$

5. $x^2y^2 - 7x^3 - 5 = 0$

6. $x^2 + y^2 = 25$

7. $3xy + 6y^2 = 15$

8. $y + xy + y^2 = 2$



Example

Find *i.* $\frac{dy}{dx}$, *ii.* $\frac{d^2y}{dx^2}$ and *iii.* $\frac{d^3y}{dx^3}$ for each of the following functions;

1. $y = 3x^4 + 2x^3 + \frac{2}{x^2}$

2. $y = x^6 + 2x^3 - 3x^2 + 4x + 1$

3. $y = x^4 - x^3 + 4x - 1$

4. $y = 3x^3(x^2 + 4)^2$

5. $y = -5 \cos 3x$

6. $y = \frac{x}{1+x^2}$



Example

Find the gradient of the curve at the indicated point

1. $f(x) = (x^2 + 3)^2$ at $x = \frac{1}{2}$

2. $y = (x - 3)(x^2 + 2)$ at $x = 1$

3. $x^2 + y^2 = 9$ at $x = 1$

4. $y = x^2 - 3$ at $x = 1$

5. $y = 3x^2 - 2$ at $x = 3$

6. $y = \sqrt{x}$ at $x = 2$

7. $y = (x - 3)(x + 4)$ at $x = -1$

8. $y = \frac{1}{x}$ at $x = 3$



Examples

1. Find the coordinates of the point on the curve $y = \frac{2}{x^2}$ at which its gradient is $\frac{1}{2}$
2. Find the values of x for which the gradient function of the curve $y = \frac{1}{4}x^4 + \frac{2}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$ is zero
3. Find the coordinate of the point on the given curve at which its gradient, m has the given value;
 - a. $y = x^2 - x + 3$; $m = 1$
 - b. $y = x^3 + x^2$; $m = 1$
 - c. $y = 5 + 3x - 2x^2$; $m = -3$
 - d. $y = (x + 3)(x - 5)$; $m = 0$
 - e. $y = (x + 1)(x - 1)$; $m = 2$



Example

Given that $y = x^3 - 4x^2 + 5x - 2$, and that P is a point on the curve at $x = 3$

1. Find $\frac{dy}{dx}$
2. Find the coordinates of P
3. Calculate the gradient of a tangent at P
4. Find the equation of the tangent at P
5. Find the equation of the normal at P
6. Find the value of x for which the curve has gradient 5



Example

Find the equation of the tangent and the normal to the curve $y = x^3 - 3x + 2$ at the point $(2, 4)$

Example

Find the coordinates of the points on the curve $y = x^3 - 4x^2 - 3x - 2$ where the tangent is parallel to the x-axis

Example

The gradient of a curve at a point $P(x, y)$ is $\frac{dy}{dx} = 4x^2 - 3x + 1$. If the curve passes through the point $A(1, 0)$, find the equation of the tangent to the curve at A



Example

Given that $y = (x - 2)(x^2 - 3x - 5)$, find $\frac{dy}{dx}$. Determine the gradient of the tangent to the curve at where it cuts the x - axis

Example

Find the gradient and the equation of the tangent and normal to the curve $y = 5x^3 - 7x^2 + 3x + 2$ at $(1, 3)$

Example

The tangent to the curve $y = x^3 + kx$, where k is a constant at $x = 1$ passes through the points $A(-1, 6)$ and $B(2, -15)$. Find the value of k

Example

Find the gradient of $\frac{x^2}{25} + \frac{y^2}{4} = 1$ at $P(3, \frac{8}{5})$



Example

A circular ink blot has a radius 2cm . At what rate is the area increasing with respect to the radius?

Example

A spherical balloon is being inflated and the volume is increasing at the rate of $15\text{cm}^3/\text{sec}$. At what rate is the radius increasing when it is 10cm ?

Example

A spherical balloon is blown up so that its volume increases at the rate of $2\text{cm}^3/\text{s}$. Find the rate of increase of the radius when the volume of the balloon is 50cm^3 .

[Hint : $V_s = \frac{4}{3}\pi r^3$, $S_A = 4\pi r^2$]



Example

A spherical balloon is blown up so that its volume increases at the rate of $2\text{cm}^3/\text{s}$. Find the rate of increase of the radius when the volume of the balloon is 50cm^3 .

$$[\text{Hint} : V_s = \frac{4}{3}\pi r^3, \quad S_A = 4\pi r^2]$$

Example

A spherical bubble is decreasing in volume at a rate of $2\text{cm}^3/\text{s}$. Find the rate at which the surface area is diminishing when the radius is 3cm . $[\text{Hint} : V_S = \frac{4}{3}\pi r^3, \quad S_A = 4\pi r^2]$



Examples

1. A spherical balloon is inflated by a gas being pumped in at a constant rate of $200\text{cm}^3/\text{s}$. What is the rate of increase of the surface area of the balloon when the radius is 100cm ?
2. An elastic spherical balloon is being blown up so that the radius is increasing at a rate of $1\text{cm}/\text{s}$. Calculate the rate at which the radius is 5cm
3. The side of a cube is increasing at a rate of $60\text{m}/\text{s}$. Find the rate of increase of the volume when the length of the side is 9cm
4. The volume of a cube is increasing at a rate of $\frac{5}{2}\text{cms}^{-1}$. Find the rate of change of the side of the base when its length is 2cm



Examples

1. The surface area of a sphere is $4\pi r^2$. If the radius of the sphere is increased from 10cm to 10.1cm , what is the approximate increase in the surface area?
2. The radius r of a circle is 5cm . Find the increase in the area A of the circle when the radius expands by 0.01cm
3. The side of a cube increases by 5% . Find the corresponding percentage increase in the volume
4. Find the approximate value for the square root of 16.01
[Let $y = \sqrt{x}$, $x = 16$, $\Delta x = 0.1$]
5. The radius of a sphere is increased by 2.5% . Find the percentage increase in the volume of the sphere
6. If $y = \sqrt{x}$, find the approximate increase in y if x is increased from 4 to 4.01



Examples

1. A curve is defined by the function
 $f(x) = x^3 - 6x^2 - 15x - 1$, find;
 - a. The derivative of $f(x)$
 - b. The gradient of the curve at the point where $x = 1$
 - c. The maximum and the minimum points
 - d. The maximum and the minimum values of $f(x)$
2. Find the maximum and the minimum points of
 $y = x^2 - 3x + 5$
3. Find the stationary points of $y = \frac{1}{3}x^3 - 2x^2 + 3x$ and identify their nature
4. The curve $y = x^2 - ax + b$ has a turning point at $(1, 3)$. Find a and b
5. The function $f(x) = ax^2 + bx + c$ has gradient function $4x + 2$ and stationary value of 1. Find the values of a , b and c



Examples

1. A rectangular sheet of metal 60cm by 50cm has squares each of side $x\text{ cm}$ cut off from its corners. The remainder is then formed into a cuboid of volume $V\text{ cm}^3$
 - a. Find the volume of x for which the volume is maximum.
 - b. Find the maximum volume
2. The length of the sides of a rectangular sheet of metal are 8cm and 3cm . A square of side $x\text{ cm}$ is cut from each corner of the sheet, and the remaining piece is folded to make an open box.
 - a. Show that the volume, V of the box is given by
$$V = 4x^2 - 22x^2 + 24x\text{ cm}^3$$
 - b. Find the value of x for which the volume is maximum and find the maximum volume.



3. A rectangular box with a square base x cm and height h cm is to be constructed from a cardboard,
- If the box holds a volume of 200cm^3 , find a formula for h in terms of x .
 - Show that the area of the cardboard needed is given by $2x^2 + \frac{800}{x}$
 - Find, to 1 d.p the least area of cardboard needed to make the box
4. A rectangle has perimeter 28m . What is its maximum area?
5. 100cm of fencing is to be used to make a rectangular enclosure. Find the greater positive area that can be enclosed.



6. A ball is thrown vertically upwards from the ground level and its height after t seconds is $h = (50t - 16t^2) \text{ m}$. Find,
- The greatest height reached
 - The time taken to reach the greatest height
7. Determine the area of the largest piece of rectangular ground that can be enclosed by 100m of fencing, if part of an existing straight wall is used as one side.
8. A cylindrical can has a radius r and a height h . The sum of the radius and the height is 2m . Find an expression for the volume, V of the cylinder in terms of r only. Hence find the maximum volume



6. Increasing and Decreasing Functions: The derivative of a function $y = f(x)$; $\frac{dy}{dx}$ gives the gradient function of the curve. If $\frac{dy}{dx} > 0$ then the function is increasing and when $\frac{dy}{dx} < 0$ the function is decreasing

Example

Find the range of values of x for which the following functions are increasing.

1. $x^2 - 2x - 5$
2. $2x^3 + 3x^2 - 12x + 5$
3. $x^3 - 3x^2 + 3x + 2$



Example

Find the interval on which $f(x) = x^3 + x^2 - 5x - 5$ is

1. Increasing
2. Decreasing

Example

Find the range of values of x for which the function

$$f(x) = x^3 - 12x + 5 \text{ is}$$

1. Increasing
2. Decreasing



Example

1. Sketch the curve $y = x^3 - 7x^2 + 15x - 9$ indicating clearly its points of intersection with the coordinate axes and its turning points.
2. Sketch the following curves
 - a. $y = x(x^2 - 16)$
 - b. $y = x^2 - 4x - 5$
 - c. $y = x^2 - 6x$
 - d. $y = 5x - x^2$
3. A function f is defined by $f(x) = 2x^3 + 3x^2$
 - a. Find the turning points and distinguish between them
 - b. Sketch the curve of $f(x)$
4. A function f is defined by $f(x) = \frac{x^3}{3} - \frac{x^2}{2} - 6x + \frac{5}{3}$
 - a. Find the turning points and distinguish between them
 - b. Sketch the curve of $f(x)$



Examples

1. A particle moves in a straight line such that its distance x meters from a fixed point O after t seconds is given by $x = t^3 - 2t^2 + t + 10$. Calculate
 - a. Its average velocity after 2 seconds
 - b. Its average velocity during the 3rd second
 - c. The average acceleration from $t = 1$ to $t = 3$
 - d. When and where the particle is momentarily at rest
2. An object is thrown into the air such that its height, h meters after t seconds is $h = 36t - 4t^2$. Find
 - a. The velocity at $t = 2$
 - b. The time taken to reach the maximum height



3. An object projected vertically upwards satisfies the relation $h = 27t - 3t^2$, where h meters is the height after t seconds.
- Find the time it takes to reach the highest point
 - How high does it go?
4. A ball thrown vertically upwards at time t has height $h = 32t - 8t^2$
- Find the velocity of the ball at $t = 1$ second
 - Find the max height it reached.
5. A particle p moves in a straight line so that its distance from a fixed point O after t seconds is $x = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$. Show that the particle is at rest at two different times and find these times. Find the acceleration of the particle at these times and interpret your results



6. A particle moves in a straight line so that its distance x from a fixed point O after t seconds where $t \leq 0$ is given by $x = 9t^2 - 2t^3$
- Find v and a at $t = 3$
 - Find x when $t = 4$ and show that the motion is towards the fixed point O .
7. A particle projected from a fixed point with velocity 10ms^{-1} , moves in a straight line such that its velocity after t seconds is $v = bt^2 + 7t + c$, where b and c are constants. 1 second after projection, its acceleration is 1ms^{-1} . Find the value of b and c .



8. A particle moves along a straight line such that its distance x from a fixed point after t seconds is
- $$x = -\frac{1}{3}t^3 + t^2 - \frac{5}{7}$$
- Find the time when the particle is momentarily at rest
 - Find the average acceleration of the particle between $t = 0$ and $t = 10$
9. A particle p moves along the x -axis such that its distance x from a fixed point O is given by $x = t^3 - 6t^2 + 9t$. Calculate
- The times at which the particle is stationary
 - The distance of p at these times
 - The acceleration of p at these times
 - The velocity of the particle when $a = 0$



10. A missile fired from the ground level rises x meters vertically upwards in t seconds and $x = 100t - \frac{25}{2}t^2$. Find
- The initial velocity of the missile
 - The time when the height of the missile is maximum
 - The maximum height reached
 - The velocity with which the missile strikes the ground
11. A particle p moves in a straight line such that displacement from a fixed point A is $x = (3t^2 + 4)$
- Find the velocity of the particle at $t = 2$
 - Find the initial displacement of p from A
12. A particle starts from rest and travels in a straight line so that its acceleration is $(t + 3)ms^{-2}$ at time t . Find the distance traveled in the interval of the time from $t = 1$ to $t = 3$



Example

1. If $x = t^2$ and $y = t^3$, find $\frac{dy}{dx}$ in terms of t and show that the above parameter representation is of the curve $y = x^{3/2}$
2. Find $\frac{dy}{dx}$ in terms of t when $x = at^2$ and $y = 2at$, where t is a parameter and a is a constant, Find the equation with this parametric representation.
3. Find $\frac{dy}{dx}$ in terms of t in each of the following cases
 - a. $x = \frac{t}{1-t}, \quad y = \frac{t^2}{1-t}$
 - b. $x = \frac{2t}{t+2}, \quad y = \frac{3t}{t+3}$
4. Find the curve whose parameter representation is given in each case in (3.)



Examples

1. A cylindrical container of thin metal is to hold 10cm^3 of water. If an engineer wishes to use the least material of metal for the construction of the cylinder, show that $A = 2\pi r^2 + \frac{20}{r}$, where r is the radius of the base area, h is the height of the cylinder and A is the surface area. Hence show that the area of the metal needed is minimum when the height of the cylinder is twice the base radius
2. A point p whose x-coordinate is α is on the line $y = 3x - 7$. If Q is the point $(4, 1)$, show that $|PQ|^2 = 10\alpha - 56\alpha + 180$, and find the value of α which makes $|PQ|^2$ minimum.



3. A sealed cylindrical can is of radius r and height h . The area of its total surface is A and its volume is V
- Find an expression for A in terms of r and h .
 - If $A = 24\pi$, find an expression for h in terms of r and hence an expression for V in terms of r
 - Find the value of r which maximises V
4. A cylinder is such that the sum of its height and the circumference of its base is 5cm . Express the volume V in terms of the radius of the base r . What is the maximum value of the cylinder?



Example

A company estimates that the cost of producing x items of a commodity is given by $C(x) = 2680 + 2x + 0.001x^2$.

1. Find the cost of producing 100 items.
2. Find the average cost of producing 200 items
3. Find the marginal cost of producing 300 items
4. At what production level will the average cost be minimum? What is the minimum average cost?



Examples

1. Determine the production level that will maximize the product for a company with respective cost and demand function $C(x) = 84 + 1.26x - 0.01x^2 + 0.00001x^3$ and $P(x) = 3.5 - 0.01x$
2. The average cost of producing x units of a commodity is $AC(x) = 21.4 - 0.002x$. Find the marginal cost at a production level of 1000 units and interpret your answer.



3. For each cost function given below, find;
- The cost, average cost, marginal cost at a production level of 1000 units
 - The production level that will minimize the average cost
 - The minimum average cost
 - $C(x) = 40000 + 300x + x^2$
 - $C(x) = 25000 + 120x + 0.1x^2$
 - $C(x) = 16000 + 200x + 4x^{3/2}$
 - $C(x) = 10000 + 340x - 0.3x^2 + 0.0001x^3$
4. For the given cost and dd function, find the production level that will maximize profit.
- $C(x) = 680 + 4x + 0.01x^2$; $P(x) = 12$
 - $C(x) = 680 + 4x + 0.01x^2$; $P(x) = 12 - \frac{x}{500}$
 - $C(x) = 1450 + 36x - x^2 + 0.001x^3$; $P(x) = 60 - 0.01x$
 - $C(x) = 16000 + 500 - 1.61x^2 + 0.004x^3$;
 $P(x) = 1700 - 7x$



Example

DBS Industry has determined that its cost of producing x IBR brand of roofing sheets can be modeled by the function

$C(x) = 2x^2 + 15x + 1500$, $0 \leq x \leq 200$, where x is the number of IBR roofing sheets produced each week and $C(x)$ represents the weekly cost in Ghana cedis. The company also determines that the price -dd function for the roofing sheets is $P(x) = -0.3x + 460$

1. Determine the profit function P for the roofing sheets
2. Determine the marginal profit function
3. Compute $MP(60)$ and $MP(145)$ and interpret the results



Example

The revenue function for the production of x boxes of roofing nails per week is $R(x) = 0.3x^2 + 460x$. Compute $R(110)$ and $MR(110)$ and interpret them

Example

Odike shoe manufacturing company knows that, for its executive shoes, the daily cost function can be modeled by the function $C(x) = 700\sqrt{x} + 500$, $0 \leq x \leq 500$, where x is the number of pairs of shoes produced daily and $C(x)$ is the daily cost in cedis

1. Determine the average cost function, $AC(x)$
2. Evaluate and interpret $C(400)$ and $AC(400)$



Example

Determine the marginal cost function $MC(x)$ and marginal profit function $MP(x)$ for each of the following;

1. $C(x) = 5x + 500$; $P(x) = 6$

2. $C(x) = 12x + 4500$; $P(x) = 15$

3. $C(x) = \frac{x^2}{100} + 7x + 1000$; $P(x) = -\frac{x}{20} + 15$

4. $C(x) = \frac{1}{100}x^2 + \frac{1}{2}x + 8$; $P(x) = 1 - \frac{x}{200}$



Rules of Integration

1. The Power Rule: For any real number n , where $n \neq -1$, the indefinite integral of x^n is $\int x^n dx = \frac{1}{n+1} x^{n+1}$

Example

Compute the following integrals

1. $\int x^8 dx$

2. $\int \sqrt{x} dx$

3. $\int x^{-5} dx$

4. $\int x^{1/3} dx$



2. Constant Rule: If k is any constant, then the indefinite integral of k with respect to x is $\int k \, dx = k \int dx = kx + c$

Example

Evaluate the following integrals

1. $\int (2x - 5) \, dx$

2. $\int 3x^2 \, dx$

3. $\int 2x^4 \, dx$



3. Sum and Difference Rule: For integrable functions $f(x)$ and $g(x)$ we have $\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$

Example

Find the following integrals

1. $\int (x^2 + 3) dx$

2. $\int (x^{1/3} + 5) dx$

3. $\int \frac{2x^3 - 3x}{4x} dx$

4. $\int (x^3 + \frac{1}{x^2} + \sqrt{x}) dx$

5. $\int (2x^2 - \frac{1}{x^2} + x) dx$



Example

Evaluate the following integrals with respect to x

1. $\int_1^3 (3x^2 + 2x + 4) dx$

2. $\int_1^2 \frac{(x+1)(x^2-2x+2)}{x^2} dx$

3. $\int_{-2}^2 (x^3 - x^2 + 1) dx$

4. $\int_{-2}^2 (x^2 - 2)(2x - 1) dx$

Example

If $\int_{-1}^a 2(x + 2) dx = 15$, where $a > 0$, find the value(s) of a .



Integration by Substitution

Functions which require integrating are not always in the standard form which can easily be integrated. However, it is possible to change them into a form which can be integrated by using an algebraic substitution.

Example

Evaluate the following integrals

1. $\int x\sqrt{1-x^2} dx$

2. $\int 3x(4x^2+3)^5 dx$

3. $\int \frac{2x}{\sqrt{4x^2-1}} dx$

4. $\int x(x^2-3)^4 dx$

5. $\int x^2(1-x^3) dx$

6. $\int (2+7x)^7 dx$

7. $\int \frac{1}{4x+5}^3 dx$

8. $\int \frac{1}{\sqrt{1-2x}} dx$



Example

Compute the following definite integrals

1. $\int_0^2 \frac{3x}{\sqrt{2x^2+1}} dx$

2. $\int_1^3 5x\sqrt{(2x^2+7)} dx$

3. $\int_0^1 x^2\sqrt{x^3+1} dx$

4. $\int_0^2 x(x^2+1)^3 dx$

5. $\int_{-1}^2 \frac{3}{(2y+3)^{1/3}} dy$

6. $\int_2^{10} \frac{1}{\sqrt{2y+5}} dy$



Type I - Denominator with Linear Factor

Example

Express $\frac{6}{(x+3)(x-3)}$ as a partial fraction

Example

Resolve the following rational functions into partial fractions.

1. $\frac{3x+2}{x^2+x-2}$

2. $\frac{11-3x}{x^2-2x-3}$

3. $\frac{9}{2x^2+x}$

4. $\frac{4x-2}{x+2x}$



Type II - Denominator with Quadratic Factor

Example

Resolve the following rational functions into partial fractions.

$$1. \frac{3x+1}{(x-1)(x^2+1)}$$

$$2. \frac{7x^2+5x+13}{(x^2+2)(x+1)}$$

$$3. \frac{x^2+1}{(x^2+2)(x-1)}$$

$$4. \frac{1}{x(x^2+1)}$$

$$5. \frac{13x}{(x-3)(x^2+x+1)}$$

$$6. \frac{x^2+9x+6}{(x^2+1)(x+1)}$$

$$7. \frac{x}{(x-1)(x^2+1)}$$

$$8. \frac{11-4x-x^2}{(x+2)(x^2+1)}$$



Type III - Denominator with Repeated Factor

Example

Express these rational functions as a partial fraction.

1. $\frac{1}{(x+2)(x-1)^2}$

2. $\frac{5x^2-2x-19}{(x+3)(x-1)^2}$

3. $\frac{2}{x^2(x-1)}$

4. $\frac{5x+4}{(1-x)(x+2)^2}$

5. $\frac{11x+12}{(x+2)(x-3)^2}$

6. $\frac{x^2-1}{x^2(2x+1)}$



Type IV - Improper Fractions

Example

Resolve these rational functions into partial fractions.

1. $\frac{2x^2+1}{(x-1)(x+2)}$

2. $\frac{x^2+1}{x^2-3x+2}$

3. $\frac{x^3+3}{(x+1)(x-1)}$

4. $\frac{x^2-2}{(x+3)(x-1)}$

5. $\frac{x^2+3}{(x-2)(x+2)}$



Examples

Compute the following integrals

$$1. \int_{-6}^{-2} \frac{1}{x} dx$$

$$2. \int_1^5 \frac{6}{x} dx$$

$$3. \int \frac{1}{x+2} dx$$

$$4. \int \frac{4x}{x^2+5} dx$$

$$5. \int \frac{1}{2x+3} dx$$

$$6. \int \frac{2x}{3x^2-2} dx$$

$$7. \int \frac{x^3}{x^4+10} dx$$



Example

Compute the following integrals

$$1. \int \frac{5}{(x-2)(x+3)} dx$$

$$2. \int \frac{4}{x^2-4} dx$$

$$3. \int \frac{2x-1}{(x+1)^2} dx$$

$$4. \int \frac{x}{4-x^2} dx$$

$$5. \int_2^3 \frac{5+x}{(1-x)(5+x^2)} dx$$

$$6. \int_2^3 \frac{x-4}{(x+2)(x-1)} dx$$



Example

Evaluate the following integrals, correct to 3 *sf*

$$1. \int_3^5 \frac{2}{x^2-1} dx$$

$$2. \int_{-1}^0 \frac{2}{(x-1)(1+x^2)} dx$$

$$3. \int_2^3 \frac{x-9}{x(x-1)(x+3)} dx$$



Example

Find the following integrals

$$1. \int \frac{x}{1+x^2} dx$$

$$2. \int \frac{x}{1+x} dx$$

$$3. \int \frac{x}{1-x^2} dx$$

$$4. \int \frac{x-2}{x^2-4x-5} dx$$

$$5. \int \frac{7x+2}{3x^3+x^2} dx$$

$$6. \int \frac{x}{16-x^2} dx$$



Integration of Exponential Functions

We recall that $\frac{d}{dx}(e^x) = e^x$. Thus $\int e^x dx = e^x + c$. In general, if $f(x)$ is a differentiable function of x which yields a constant upon differentiation then

$$\int e^{f(x)} dx = \frac{1}{f'(x)} \cdot e^{f(x)} + c. \text{ Also } \int e^{kx} dx = \frac{1}{k} \cdot e^{kx} + c$$

Examples

1. $\int e^{2x} dx$

2. $\int (2 - e^{2x}) dx$

3. $\int 10xe^{x^2} dx$

4. $\int (2x + 1)e^{2x^2+2x} dx$

5. $\int e^{2x+3} dx$

6. $\int x^2 e^{x^3} dx$



Example

Determine the average value of $f(x) = x^3 - 2x^2 - 5x + 1$ on the interval $[-2, 4]$

Example

The annual per capita consumption of alcoholic beverages in the U.S. can be modeled by the function

$f(x) = 43e^{-0.012x}$, $1 \leq x \leq 11$, where x is measured in years. Determine the average per capita consumption of alcoholic beverages



Examples

1. Determine the area enclosed by $y = 2x + 3$, the x-axis and the ordinates $x = 1$ and $x = 4$
2. Find the area enclosed by the curve $y = 3x + 4$, the x-axis and the lines $x = 1$ and $x = 4$
3. Determine the area bounded by the curve $y = x^2 + 5$, the x-axis and the lines $x = 0$ and $x = 3$
4. Sketch $y = x^2 - 4x - 5$ and find the area bounded by the curve and the x-axis
5. Determine the area enclosed between the curve $y = x(x - 1)(x - 2)$ and the x-axis



3. The Area between two curves: The area enclosed between two curves $y = f_1(x)$ and $y = f_2(x)$ is given by

$$A = \int_a^b f_2(x) dx - \int_a^b f_1(x) dx$$

Examples

1. Sketch the curves $y = x^2 + 1$ and $y = 7 - x$ on the same axis and find the area enclosed between the curves
2.
 - a. Determine the coordinate of the points of intersection of the curves $y = x^2$ and $y^2 = 8x$
 - b. Sketch the curves $y = x^2$ and $y^2 = 8x$ on the same axis
 - c. Calculate the area enclosed by the two curves
3. On the same axis, sketch and label clearly the curves $y = x^2 - 5$ and $y = 5 - x^2$. Find the area of the finite region enclosed by the two curves



Examples

1. A curve passes through the point $(1, 0)$. If the gradient function at any point $P(x, y)$ is $3x^2 - 1$, find the equation of the curve.
2. The gradient of a curve at any point $P(x, y)$ is $4x^2 - 3x + 1$. If the curve passes through the point $A(1, 0)$, find the equation of the curve.
3. The gradient of a curve at any point is given by $(2x - 3x^2)$. If the curve passes through the point $(1, 2)$, find the equation of the curve.
4. A curve passes through the point $(1, -11)$ and its gradient at any point is $ax^2 + b$, where a and b are constants. The tangent to the curve at the point $(2, -16)$ is parallel to the x-axis. Find;
 - a. The values of a and b
 - b. The equation of the curve



Examples

1. The demand function for a product is $p = 100 - 0.5q$, where p is the price and q is the quantity demanded. If the supply function is $p = 10 + \frac{q}{10}$, determine the consumers surplus
2. In each of the cases below, determine the total benefit/gain to consumers who are willing to pay more than the equilibrium price

a. $p_d = 20 - 0.8q$

$$p_s = 4 + 1.2q$$

b. $p_d = 900 - q^2$

$$p_s = 100 + q^2$$

c. $p_d = \frac{50}{q+5}$

$$p_s = \frac{q}{10} + 4.5$$

d. $p_d = 400 - q^2$

$$p_s = 200q + 100$$

e. $q = 100(10 - p)$

$$q = 80(p - 1)$$

f. $q = \sqrt{100 - p}$

$$q = \frac{p}{2} - 10$$



Let p_0 and q_0 be the equilibrium price and the quantity supplied respectively. The shaded area depicts the total gain to producers. If the supply function is $p = f(q)$, then producers surplus (PS), is given by

$$P.S = p_0 q_0 - \int_0^{q_0} f(q) dq$$

Example

The demand function for a product q is $p = \frac{90}{q+2}$ and the supply equation is $p = q + 1$. Determine the consumer's surplus.



Examples

1. Given the demand function $P = 42 - 5Q - Q^2$ and assuming that the equilibrium price is GH¢6. Find C.S
2. Find the C.S when equilibrium price and quantity are respectively GH¢32.5 and GH¢25 if $P = 45 - 0.5Q$
3. Given that $P = (Q + 3)^2$, find the P.S at $P_0 = 81$ and $Q_0 = 6$
4. Given the demand function $P = 25 - Q^2$ and the supply function $P = 2Q + 1$, assuming that there is pure competition in the economy, find
 - a. CS
 - b. PS



Exercise

1. Given the demand function $P = 113 - 3Q^2$ and the supply function $P = (Q + 1)^2$ under pure competition, find CS and PS.
2. Under monopoly, the quantity sold and market prices are determined by the demand function. If the demand function for a profit maximizing monopolist is $P = 274 - Q^2$ and $MC = 4 + 3Q$. find CS



Example

Find the general solution of the following differential equations

1. $\frac{dy}{y} = 3x^2$

2. $\frac{dy}{y} = \frac{x^2}{y^3}$

3. $\frac{dy}{y} = 2x$

4. $\frac{dy}{y} = xy$

5. $x^2 \frac{dy}{y} = y(y - 1)$

6. $\frac{dy}{y} = \frac{x}{y}$

7. $\frac{dy}{y} = \frac{y}{x}$

Example

Determine the particular solution for the following differential equations

1. $\frac{dy}{dx} = 3xy; \quad (x = 0, y = 3)$

2. $\frac{dy}{dx} = xy - 3x; \quad \text{at } y(0) = 1$



Example

The annual rate of increase in employees between 1950 and 1985 at the NewMedia Company can be modeled by the differential equation $\frac{dy}{dx} = 0.3t^{0.6}$, $0 \leq t \leq 35$ where t represents the number of years and y represents the number of employees in 100s

1. If the company had 1800 employees in 1965, find the employee growth function for the company
2. Use your answer in (1) to estimate the number of employees at the company in 1983



Example

The StrikeNow Company determines that the marginal cost for their MaxiPak of lighters is given by the differential equation $\frac{dC}{dx} = -0.02x + 6$, $0 \leq x \leq 300$ where x is the number of MaxiPaks produced and C is the cost in dollars

1. Determine the general solution of the cost function C
2. Find the particular cost function, if the cost of producing 10 Paks is 400



Exercise

Determine the general solution of the following differential equations

1. $\frac{dy}{dx} = -2$

2. $\frac{dy}{dx} = e$

3. $\frac{dy}{dx} = 1 - 3x$

4. $\frac{dy}{dx} = \sqrt{x} + 2$

5. $\frac{dy}{dx} = \frac{5}{x^2}$

6. $\frac{dy}{dx} = 10x^4 - x^2 + 3$



Exercise

Determine the particular solution of the following differential equations

1. $\frac{dy}{dx} = 12x; \quad x = 1, y = 8$

2. $\frac{dy}{dx} = 3x^2 + 4x; \quad y(1) = 6$

3. $\frac{dy}{dx} = xe^{x^2}; \quad y(0) = 2$

4. $\frac{dy}{dx} = x^3 - 2x; \quad y(0) = -2$



Examples

1. Consider $f(x, y) = 2x^2 - 3y^3$. Determine the following
 - a. $f(2, 1)$
 - b. $f(-1, 2)$
 - c. $f(1, -2)$
 - d. $f(2, 3)$
2. The cost of producing x tennis balls and y volleyballs is given by $C(x, y) = 0.97x + 0.89y$
 - a. Determine the domain of $C(x, y)$
 - b. Evaluate $C(250, 700)$ and interpret it



Examples

1. Determine $f_x(x, y)$ and $f_y(x, y)$ for $f(x, y) = x^y - y^2$
2. Compute the partial derivatives for
 $f(x, y) = x^2y + y^3x - 2xy = y$
3. Determine f_x and f_y for each of the following
 - a. $f(x, y) = 5x^2 - 6y^3$
 - b. $f(x, y) = 2xy - y^2 + 1$
 - c. $f(x, y) = x^3y^2$
 - d. $f(x, y) = x^2y^4$
 - a. $f(x, y) = \frac{x}{y}$



Example

1. Donyma Steel Complex spends x thousand cedis each week on newspaper advertising and y thousand cedis each week on radio advertising. The company has weekly sales, in terms of thousands of cedis, given by $S(x, y) = 2x^2 + y$. Determine
 - a. $S_x(x, y)$ and $S_y(x, y)$
 - b. $S_x(3, 5)$ and $S_y(3, 5)$ and interpret each
2. If $S(x, y) = 3x + 2y^3$, compute $S_x(2, 6)$ and $S_y(2, 6)$ and interpret each
3. For $f(x, y) = x^2 + y^2 - 8x + 2y + 7$, determine the value of x and y such that $f_x = 0$ and $f_y = 0$ simultaneously



2. Marginal Productivity of Labour and Capital If L represents the units of labour and K represents the units of capital, then the total production Q is given by the Coff-Douglas production function

$$Q(L, K) = AL^aK^b, \text{ where } A, a \text{ and } b \text{ are constants}$$

Example

If $f(x, y) = 4x^{0.75}y^{0.25}$, determine the realistic domain of $f(x, y)$ and evaluate $f(x, y)$ when $x = 50$ and $y = 60$ and interpret if



Example

The production function for a company is given by

$f(x, y) = Q \times 1.64x^{0.64}y^{0.4}$, where x is the number of labour hours in hundreds and y is capital in thousands, and Q is the quantity in 1000s of gallons

1. Compute f_x and f_y
2. Find $f_x(100, 18)$ and interpret it
3. Find $f_y(110, 20)$ and interpret it



Remark

For some amounts invested in labour and in capital, we wish to know whether production would increase more rapidly if additional resources were invested in labour or in capital. We can approximate the change in productivity by any of the following

1. The change in productivity with respect to labour $= f_x \Delta x$
2. The change in productivity with respect to capital $= f_y \Delta y$

Example

Suppose that a production function is given by $f(x, y) = 4.23x^{0.37}y^{0.66}$, where x =labour (million) and y =capital (million). Currently, $x = 5$ and $y = 1$. Would production increase more by spending additional GH¢1m on labour or GH¢500000 on capital equipment

